Mercer Emergency Department
CariSurg MedTech Pathways

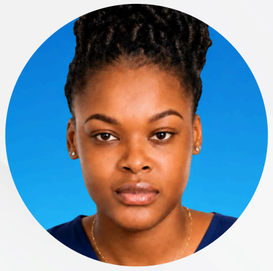
THE CLINICAL GUARDRAIL

AI-Assisted Emergency Triage for Mercer General

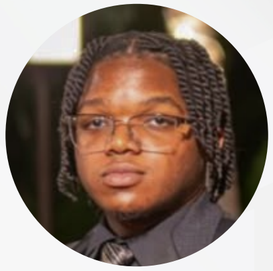
PRESENTED BY ALEKSANDRAS GROUP



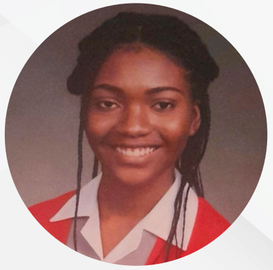
Ansarah Mohammed



Shakada Blake



Aiden Daniel

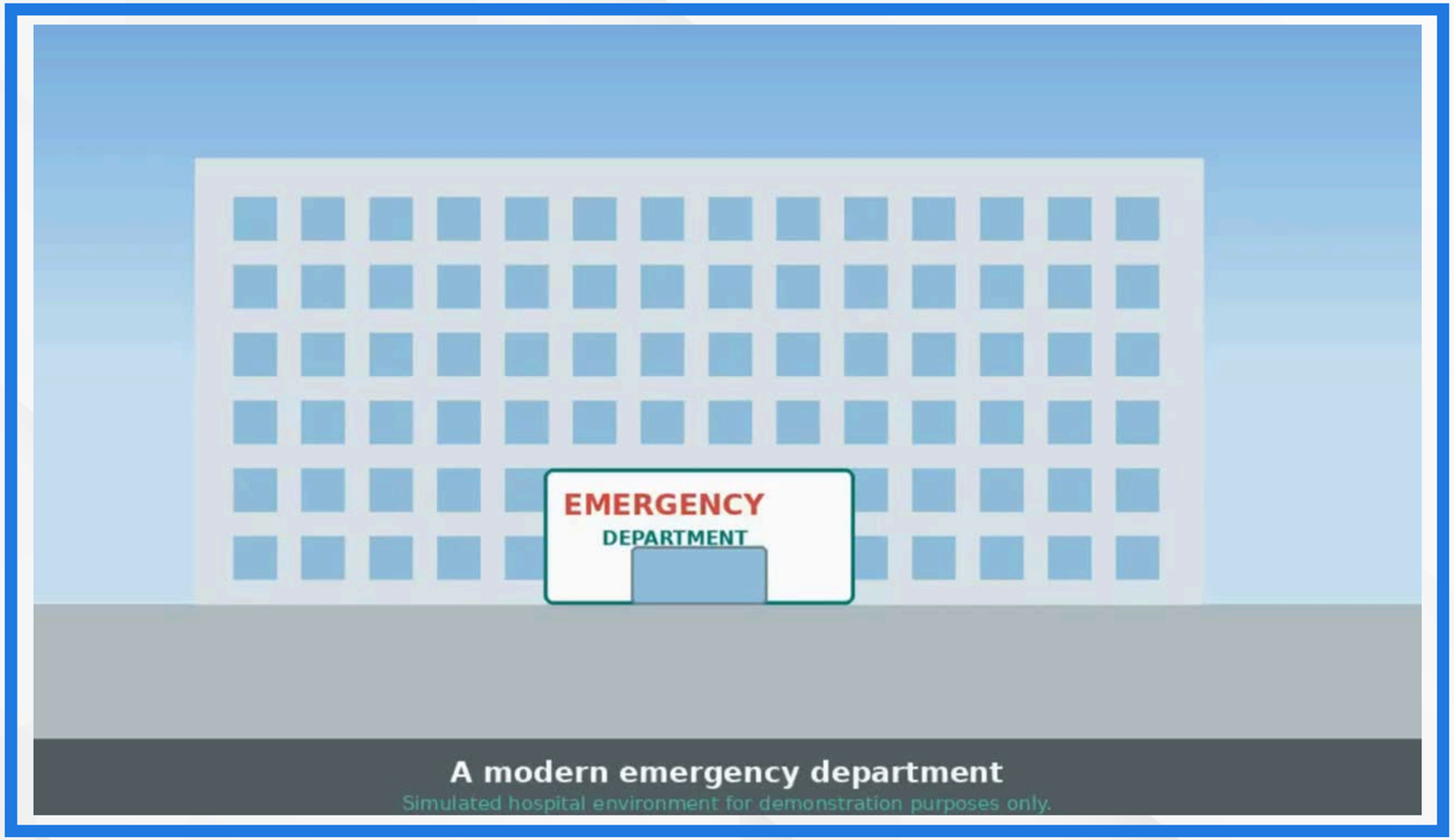


Amanda Miller

Ashi Baldie & Aryanna Boyce



FOLLOW THE PATIENT JOURNEY



ARRIVE

SENSE

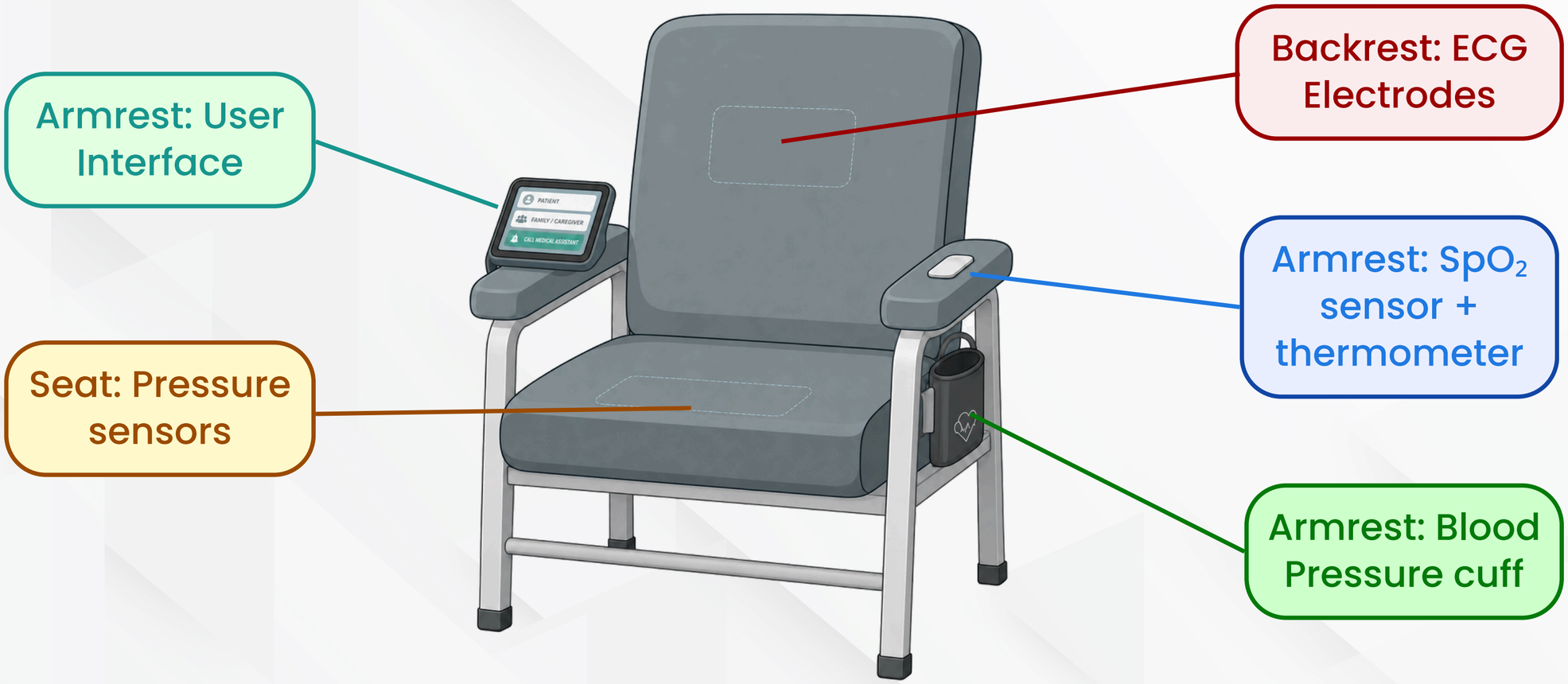
CONNECT

MONITOR

CARE

FROM ARRIVAL TO CLINICAL DECISION

CAN A CHAIR REALLY DO THIS?



RESEARCH SUPPORTS TECHNICAL FEASIBILITY - INTEGRATED CONCEPT REQUIRES CLINICAL VALIDATION

Baek et al., 2012; Hou et al., 2018; Sifuentes et al., 2019; Odesola et al., 2024

THE MODEL SUPPORTS - IT DOES NOT DECIDE

INITIAL BENCHMARK

0.2215
MACRO F1

54.5%
ACCURACY

INITIAL BENCHMARK

0.508
MACRO F1

68.3%
ACCURACY

PUBLISHED EVIDENCE

75.7 %
KATE ML ACCURACY

59.8 %
TRIAGE NURSE ACCURACY

57.8% → 64.9%
TRIAGE ACCURACY AFTER DECISION SUPPORT

NURSE FIRST. AI SECOND. CLINICIAN FINAL.

THE MODEL SUPPORTS - IT DOES NOT DECIDE

REVISED SCOPE

ESI 1
Clinician Pathway

&

ESI 2 - 5
ML Evaluation

DEPLOYMENT DECISION

Today: Shadow Mode

Nurse Makes every decision

Next: Supervised ESI 2- 5 Support

Nurses can override/accept recommendation

NURSE FIRST. AI SECOND. CLINICIAN FINAL.

THE CHAIR IS AN OPTION, NOT A BARRIER TO CARE

PATIENT ARRIVES

↓

Can the patient safely use the chair?

↙ ↘



SMART-CHAIR PATH

Guided check-in

Vital signs

Assistance button

SUPPORTED MANUAL PATH

Wheelchair or stretcher

Opt out or needs help

Chair or sensor unavailable



↘ ↙

NURSE ASSESSMENT

↓

Monitoring and reassessment



VISIBLE EMERGENCY SIGNS → IMMEDIATE CLINICIAN

WHAT COULD STILL GO WRONG?

WE DESIGNED AROUND THE THREE BIGGEST RISKS

UNRELIABLE READINGS

↓

FLAG & VERIFY

PATIENT DETERIORATION

↓

MONITOR & REASSESS

AUTOMATION BIAS

↓

ASSESS FIRST & OVERRIDE

GUARDRAILS REDUCE RISK BUT CLINICAL JUDGEMENT REMAINS ESSENTIAL.

CARIBBEAN DEPLOYMENT REQUIRES LOCAL RULES

FIVE LEGAL CONTEXTS

LOCAL LAW 1

TRINIDAD & TOBAGO

Data Protection Act,
2011

LOCAL LAW 2

GRENADA

Data Protection Act,
2023

LOCAL LAW 3

BARBADOS

Data Protection Act,
2019

LOCAL LAW 4

SAINT LUCIA

Data Protection Act,
Cap. 8.18

LOCAL LAW 5

JAMAICA

Data Protection Act,
2020

ONE SYSTEM. FIVE LEGAL CONTEXTS. COUNTRY-SPECIFIC APPROVAL BEFORE DEPLOYMENT.

CARIBBEAN DEPLOYMENT REQUIRES LOCAL RULES

THESE LAWS DIFFER IN SCOPE AND IMPLEMENTATION, SO DEPLOYMENT REQUIRES COUNTRY-SPECIFIC LEGAL REVIEW.

MINIMISE DATA

PROTECT ACCESS

REVIEW BEFORE PILOT

WHO AI - FOR HEALTH GUIDANCE

1

SAFETY & EFFECTIVENESS

2

TRANSPARENCY & ACCOUNTABILITY

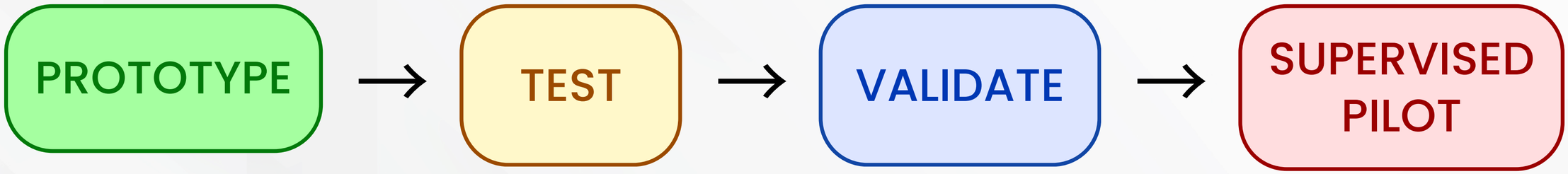
3

MONITORING THROUGHOUT DEPLOYMENT

ONE SYSTEM. FIVE LEGAL CONTEXTS. COUNTRY-SPECIFIC APPROVAL BEFORE DEPLOYMENT.

FROM CONCEPT TO CLINICAL REALITY

OUR STAGED PATHWAY



NEXT: BUILD & TEST INTEGRATED PROTOTYPE

SENSORS INFORM. AI ASSISTS. CLINICIANS DECIDE.

ANY QUESTIONS?

THE CLINICAL GUARDRAIL

AI-Assisted Emergency
Triage for Mercer General

